

INTISAT – ISM PAYLOAD

Complete Payload Architecture and Mission Engineering Document

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Classification: Engineering Working Document

1 Executive Summary

The INTISAT Image Sensor Microscopy (ISM) payload is a CubeSat-class in-situ computational microscope implementing Fourier Ptychographic Microscopy (FPM), a computational imaging technique that reconstructs a high-resolution, complex-valued image (amplitude plus phase) from a sequence of lower-resolution intensity measurements taken under angularly diverse illumination.

The concept is validated at bench level. The software pipeline (capture → FPM reconstruction → super-resolution via RealESRGAN) runs on a Raspberry Pi 4 and produces demonstrably enhanced images. The repository contains approximately 18 GB of real experimental data, production-grade Python code, and extensive sensor/algorithm documentation. No hardware design files (PCB, schematic, or Gerber) exist in the repository.

This document closes the gap between the bench-level demonstrator and a flight-credible CubeSat payload by:

- defining the hardware architecture and board partitioning,
- providing mission-level parameter tables and budgets,
- proposing an onboard versus ground processing split,
- identifying the differentiators with respect to prior work, and
- specifying the validation and test plan required for credibility.

Recommended architecture: Raspberry Pi Compute Module 4 (CM4) on a custom carrier board, with a separate sensor board and a microcontroller supervisor (Option D).

2 Repository Audit

2.1 Audit Summary

Category	Status	Evidence
Capture code	Production-ready	<code>capturaclaude.py</code> , <code>captura_circulo_movil_tftcolorSBARRIDO.py</code>

FPM reconstruction	Production-ready	<code>fpm_procesador_tft.py</code> , <code>fpm_procesador_tft_fixed.py</code> , <code>fpm-py/</code> library
Super-resolution (ESR-GAN)	Implemented / tested	<code>run_esrgan_fpm.py</code> , <code>rrdbnet_standalone.py</code>
Metadata system	Implemented	JSON schema in capture scripts
Sensor documentation	Comprehensive	<code>CMOS_SENSOR_OV5647_TECHNICAL_GUIDE.md</code>
Algorithm documentation	Comprehensive	<code>TECHNICAL_REPORT_FPM_WORKFLOW.md</code>
Real experimental data	Present (~18 GB)	<code>data/raw/</code> , <code>data/processed/</code> , <code>data/calibration/</code>
Hardware design (PCB/SCH)	Missing	Directory stubs only, no KiCad/Eagle/Gerber
Bus subsystems	Stubs only	ADCS, EPS, OBC, TCS all <code>.gitkeep</code>
Ground segment software	Stub	<code>ground/mcs/</code> , <code>ground/station/</code> empty
Link budget simulation	Stub	<code>sim/link_budget/</code> empty
Orbital dynamics sim	Stub	<code>sim/dynamics/</code> empty
Power budget	Missing	No power analysis found
Thermal analysis	Missing	No thermal model found
Calibration data	Sparse	<code>data/calibration/</code> minimal content

2.2 What Already Exists

- Complete FPM acquisition software (controlled, swept, circular illumination patterns).
- Full FPM reconstruction pipeline including NaN-safe variant.
- RealESRGAN 4× super-resolution integrated.
- OV5647 sensor interface via `picamera2` over MIPI CSI-2.
- OLED illumination control via I2C (SSD1306, 128×64).
- Real experimental data sets and JSON metadata schema.
- Calibration framework documented, though data remain sparse.

2.3 What Is Missing

- Hardware design files (schematics, PCB, BOM).
- Power budget and conditioning design.
- Thermal analysis.
- Mechanical integration drawings.
- Communications/downlink software.
- Ground station software.
- Orbital or mission simulation artifacts.

- Radiation tolerance assessment.
- Watchdog/supervisor firmware.
- Flight software architecture and qualification plan.

3 System Objective

The INTISAT ISM payload shall demonstrate computational lensless microscopy from low Earth orbit by:

1. acquiring raw intensity image sequences under controlled angular illumination patterns using an OLED or LED array at a fixed distance from a bare CMOS sensor,
2. reconstructing a high-resolution complex-field image on-orbit using Fourier Ptychographic Microscopy,
3. downlinking reconstructed science products and selected raw/diagnostic data,
4. optionally applying super-resolution onboard or on the ground, and
5. demonstrating the viability of computational imaging as a space instrument modality.

Primary objective: demonstrate that a sub-100 g, sub-3 W computational microscope payload can produce sub-micron-equivalent imagery of biological, material, or optical samples in LEO, with onboard reconstruction and data compression.

4 Mission Context and Assumptions

Parameter	Value	Justification / Source
Orbit type	LEO Sun-Synchronous	Stable thermal environment, regular ground contact
Altitude	500–550 km	Typical CubeSat deployment altitude
Inclination	~97.5°	SSO at 500 km
Orbital period	~95 min	Derived from altitude
Ground passes/day	4–6 useful	Typical LEO contact geometry
Mission duration	12 months	Minimum demonstration mission
Platform	3U CubeSat	10×10×30 cm
Payload mass allocation	≤300 g	Assumption
Payload power allocation	≤5 W peak, ≤3 W avg	Assumption
Downlink band	UHF/VHF or S-band	TBD by bus architecture
Downlink rate	9.6 kbps to 1 Mbps	Assumption; drives data budget

5 Payload Functional Description

The payload operates according to the following chain:

1. Command reception from OBC.
2. Sample readiness check.
3. Illumination sequencing over a 9×9 angular grid.
4. Raw image acquisition per angle with metadata logging.
5. Onboard preprocessing: dark subtraction, flat-field correction, bad-pixel masking.
6. FPM reconstruction into a high-resolution complex-field image.
7. Optional super-resolution stage.
8. Compression, packetization, storage, and downlink.

One complete acquisition session consists of 81 raw frames, one reconstructed image, optional super-resolved output, and metadata.

6 Subsystem Decomposition

6.1 Optical Subsystem

Field	Value
Purpose	Define lensless imaging geometry; sample placed directly above sensor
Inputs	Sample, illumination wavefront
Outputs	Diffacted intensity pattern on sensor plane
Design drivers	Working distance (8 mm), sample proximity, absence of objective lens
Dependencies	Mechanical integration, fixed z-distance
Risks	Contamination, stray light, z-drift due to thermal expansion
Still to design	Flight-compatible sample holder, dust protection, thermal compensation

6.2 Image Sensor Subsystem

Field	Value
Sensor	OmniVision OV5647, 1/4" CMOS, 5.04 MP
Resolution	2592×1944 px, $1.4 \mu\text{m}$ pitch
Interface	MIPI CSI-2, 84 MHz pixel clock
Controlled via	picamera2 with manual exposure/gain
Major risks	Radiation damage, thermal noise, hot-pixel accumulation

Current status	Bench-tested and operational
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6.3 Illumination Subsystem

Field	Value
Current device	SSD1306 OLED (bench only)
Interface	I2C at 0x3C, 400 kHz
Emission	Approximately 530 nm
Pattern	9×9 grid, 81 positions
Major risk	OLED vacuum incompatibility and outgassing
Recommendation	Replace with discrete LED matrix for flight version

6.4 Compute Subsystem

Field	Value
Recommended compute core	Raspberry Pi Compute Module 4
Processor	BCM2711, quad-core Cortex-A72 @ 1.5 GHz
RAM	4 GB LPDDR4
Storage	32 GB eMMC
Software stack	Python 3.11, numpy, scipy, picamera2, fpm-py, optional PyTorch
FPM processing time	5–8 min for 81 frames
Active power	~3.5–4.5 W
Key risks	SEU, thermal throttling, OS corruption

7 Hardware Architecture Options

Option	Description	Verdict
A	Full Raspberry Pi 4 board as-is	Rejected: excess connectors, poor CubeSat integration, weak novelty
B	CM4 + custom carrier board	Strong candidate: compact, tailored interfaces, better integration
C	CM4 + distributed support boards	Viable but more complex for first implementation
D	MCU supervisor + CM4 hybrid	Recommended: low-power standby, watchdog, stronger flight credibility

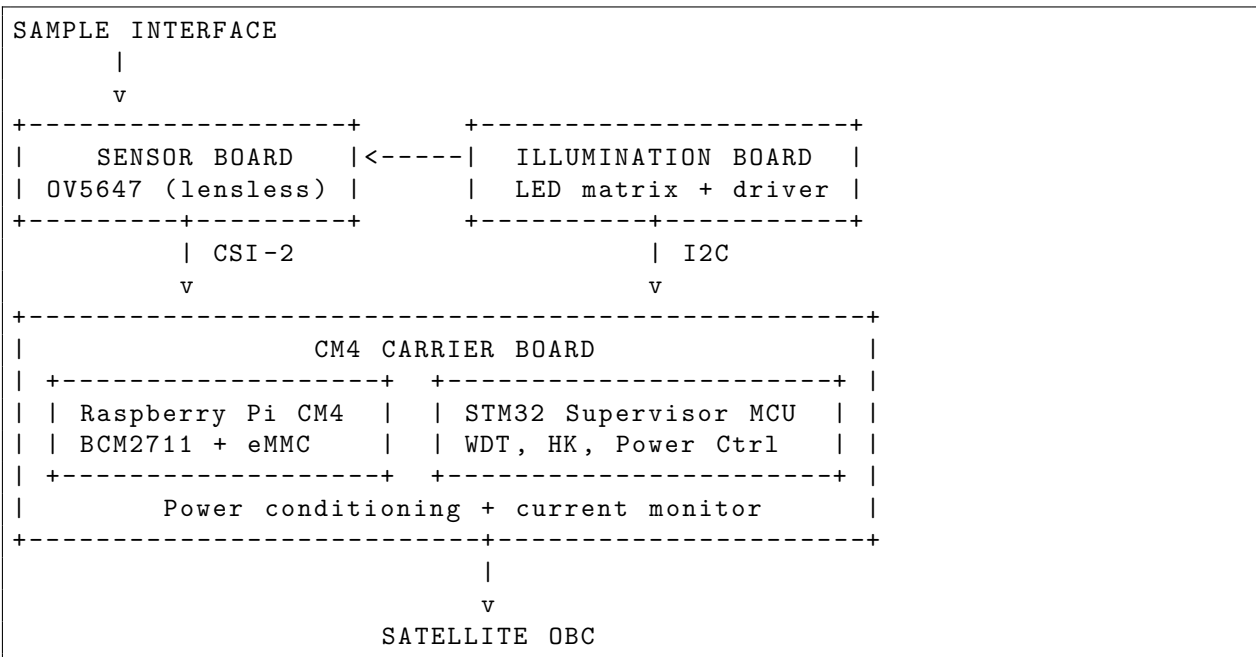
8 Recommended Architecture

The recommended approach is **Option D**: a hybrid architecture with a low-power supervisor MCU and a CM4 compute core.

- The MCU handles watchdog, housekeeping telemetry, power gating, and OBC interface.
- The CM4 powers on only during acquisition and processing windows.
- This reduces thermal load and average power consumption.
- It also makes the payload more defensible as a real CubeSat subsystem rather than a repackaged development board.

9 Block Diagrams

9.1 Overall Payload Architecture



9.2 Data Flow

```

LED pattern -> OV5647 capture -> TIFF + JSON metadata -> eMMC storage
-> dark subtraction + flat field
-> FPM reconstruction
-> reconstructed image / phase image
-> compression + packetization
-> OBC interface -> RF downlink -> Ground station

```

10 Imaging Chain Definition

Parameter	Value	Notes
Sensor	OV5647	Raspberry Pi Camera Module v2 compatible
Full resolution	2592×1944 px	5.04 MP
Pixel size	1.4 μm	Physical pitch

Bit depth	10-bit native, stored as 16-bit TIFF	Raw Bayer
Exposure range	1 μ s – 6000 ms	Software controlled
Default exposure	150 ms	From capture script
Working distance	8.0 mm	Illumination-to-sensor
Illumination wave-length	530 nm	Green dominant
FPM output resolution	4 $\times$ linear enhancement	Approx. 0.35 μ m equivalent
Frames per session	81	9 $\times$ 9 grid
Acquisition time	3–5 min	Includes settle delay and overhead
FPM reconstruction time	5–8 min	CM4 CPU-bound estimate

11 Communications and Protocol Selection

Bus	Connection	Protocol	Rate	Justification
Sensor $\rightarrow$ CM4	OV5647 to BCM2711	MIPI CSI-2	84 MHz	Native high-speed camera interface
CM4 $\rightarrow$ LED driver	BCM2711 to LED driver	I2C	400 kbps	Simple pattern control
CM4 $\leftrightarrow$ STM32	Compute to supervisor	UART	115200 baud	Sufficient for commands/HK
STM32 $\rightarrow$ OBC	Payload to spacecraft bus	CAN	250 kbps	Fault-tolerant, standard CubeSat choice

12 Data Budget Tables

Item	Count	Raw Size	Compressed Size
Raw TIFF frame	81	10.1 MB each ($\sim$ 818 MB/session)	$\sim$ 2 MB PNG each ($\sim$ 162 MB/session)
Dark frame	1	10.1 MB	$\sim$ 2 MB
Metadata JSON	81 + 1	$\sim$ 344 KB total	—
FPM reconstructed image	1	$\sim$ 45 MB TIFF	$\sim$ 2 MB JPEG2000
Phase image	1	$\sim$ 45 MB	$\sim$ 5 MB
Preview thumbnail	1	$\sim$ 512 KB	$\sim$ 100 KB

13 Power Budget Tables

Component	Standby	Active	Peak	Notes
BCM2711 / CM4	0 W	3.5 W	4.5 W	Powered off in standby
OV5647 sensor	0.01 W	0.2 W	0.3 W	Low-power CMOS
LED array	0 W	0.07 W	0.12 W	One active LED example
STM32 supervisor	0.01 W	0.05 W	0.1 W	Always available
Power conversion losses	0.03 W	0.8 W	1.0 W	Buck/LDO losses
Total	~0.07 W	~5.3 W	~7.1 W	

14 Calibration and Metrology Plan

Calibration Type	Method	Frequency	Storage	Notes
Dark frame	Same T, exposure, gain; LED off	Before each session + daily	eMMC	Must track sensor aging
Flat field	Uniform sample at each angle	Pre-launch, monthly	eMMC	Critical for quantitative imaging
PRNU	Uniform illumination, multiple exposures	Pre-launch	Lookup table	Fixed sensor characteristic
Thermal characterization	Dark frames vs temperature	Pre-launch TVAC	Lookup table	Temp-compensated correction

15 Validation and Verification Plan

Test ID	Objective	Method	Expected Output	Pass Criterion
OPT-01	Verify lensless working distance	Known target at $z = 8$ mm	Sharp diffraction pattern	FOV within ± 0.5 mm
SEN-01	Sensor verification	Functional camera test	Stable stream + control	All modes functional
CAL-01	Dark frame calibration	Average 50 dark frames	Master dark file	Stable noise floor
CAL-02	Flat-field calibration	Uniform illumination over all angles	81 flat frames	Uniformity >95%
IQ-01	FPM quality	Reconstruct known target	Reconstructed image	Meets target resolution criterion
OBP-01	End-to-end onboard chain	Capture to packetization test	Decodable science packet	Image quality acceptable

THM-01	Thermal-vacuum operation	Full session at hot/cold extremes	Functional output	Completes successfully
VIB-01	Launch vibration	Standard vibration test	Pre/post function check	No failures

16 Comparison Against Prior Work

- Many educational CubeSats fly cameras and downlink raw photos.
- PhiSat-class systems do onboard filtering/compression for Earth observation.
- Biology experiments in orbit usually rely on conventional microscopy or larger ISS-class platforms.
- Ground-based FPM exists, but the current differentiator is onboard FPM reconstruction in orbit for a lensless payload.

17 Novelty Statement

First demonstration of Fourier Ptychographic Microscopy executed entirely onboard a CubeSat in low Earth orbit, producing super-resolved complex-field microscopy images without an objective lens, using a custom carrier board based on the Raspberry Pi Compute Module 4.

18 Risks and Failure Points

ID	Description	Likelihood	Severity	Mitigation
R-01	OLED vacuum incompatibility / outgassing	High	Critical	Replace OLED with LED matrix
R-02	BCM2711 SEU during reconstruction	Medium	High	MCU watchdog, scrubbing, safe software
R-03	OV5647 radiation degradation	High	Medium	In-flight dark-frame updates
R-04	z-distance thermal drift	Medium	High	Low-CTE spacer + calibration
R-05	eMMC corruption	Medium	High	Journaling, redundancy, safe writes
R-10	No hardware design files exist yet	Certain	Critical	Start carrier PCB immediately

19 Next Engineering Actions

- Define OBC interface (CAN vs UART, packet format, command set).
- Select the flight illumination element (LED matrix strongly preferred).
- Define EPS interface and available current/voltage.
- Start CM4 carrier schematic in KiCad.

- Design sensor board and illumination board.
- Adapt acquisition code for headless flight operation.
- Implement telemetry/packetization and supervisor firmware.

Appendix A — Draft Requirements

Req ID	Category	Requirement
PAY-REQ-001	Functional	The payload shall acquire at least 49 raw intensity images per session under angularly diverse illumination.
PAY-REQ-002	Functional	The payload shall reconstruct a high-resolution complex-field image onboard using FPM.
PAY-REQ-004	Interface	The payload shall interface with the OBC via CAN bus at 250 kbps.
PAY-REQ-006	Power	Average payload power shall not exceed 3 W at nominal duty cycle.
PAY-REQ-008	Mass	Total payload mass shall not exceed 250 g.
PAY-REQ-009	Volume	The payload shall fit within 1U (90×90×90 mm).
PAY-REQ-012	Reliability	The payload shall include a hardware watchdog capable of resetting the CM4 if no heart-beat is received within 30 s.
PAY-REQ-015	Calibration	The payload shall execute a dark-frame calibration sequence on command.